Power Credits & Commercial Intelligence

Financial Mechanism / Capital Rail

Connecting conventional capital with verified energy, productive capacity, and measurable commercial outcomes.

1 kWh = 1 Power Credit

KYE (Know Your Energy) | KYP (Know Your Productivity)

THE CENTRAL IDEA

Rheo connects the physical economy to financial participation: measure energy, verify productive activity, account for value, mobilise capital, and settle through defined commercial arrangements.

CORE OPERATIONAL CYCLE

1. MEASURE	2. VERIFY	3. ACCOUNT	4. FINANCE	5. SETTLE
Physical energy data	Evidence & eligibility	Credits & ownership	Capital deployment	Contractual payment

A shared infrastructure for energy evidence, commercial accounting, and capital flows with distinct rules for physical delivery, credit ownership, and financial settlement.

01 / POWER CREDIT MECHANISM

From physical energy to a digital credit: 1 Power Credit (PC) = 1 kWh of eligible, verified energy. Power Credits provide a standard accounting unit linking physical energy activity to digital records and financial transactions. They are not automatically money, a carbon credit, or a guaranteed financial return.

AUTHORISE	MEASURE	VERIFY	ISSUE
Asset, parties & ceiling	Authenticated meter data	Eligibility & integrity checks	Smart-contract issuance

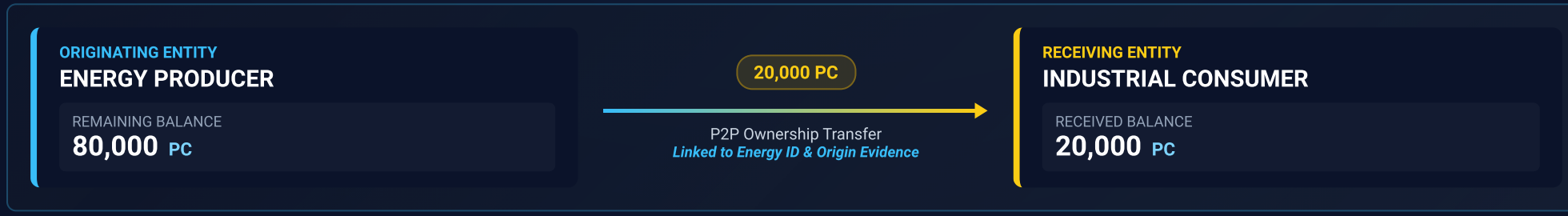
WHAT EACH STAGE ESTABLISHES

STAGE	OPERATIONAL RULE & PROOF ESTABLISHED
AUTHORISE	Register the asset, responsible parties, contractual conditions, and maximum issuance ceiling.
MEASURE	Capture energy data through authenticated meters and timestamped readings.
VERIFY	Validate asset identity, data integrity, meter accuracy, eligible quantity, and issuance history.
ISSUE	A smart contract enforces approved conditions and issues credits to the authorised wallet.

Issuance Conversion: 100,000 kWh verified → 100,000 Power Credits
The issuance ceiling is the maximum permitted quantity, not an automatic entitlement. Smart contracts enforce approved rules; they do not independently prove that physical energy was generated.

02 / OWNERSHIP & SETTLEMENT (P2P TRANSFER)

01. ISSUED	02. TRANSFER	03. REDEEM	04. RETIRE / BURN
Credit enters wallet	Ownership updates	Eligible claim applied	Removed from circulation



THREE SEPARATE LAYERS

- **PHYSICAL ENERGY:** Measured electricity in kWh.
- **POWER CREDITS:** Verified energy quantity and recorded wallet ownership.
- **FINANCIAL VALUE:** Price or contractual value assigned under an agreement.

IMPORTANT DISTINCTION

P2P transfer represents ownership of credits, not necessarily physical electricity delivery. Physical supply remains subject to separate grid, commercial, and contractual arrangements.

RHEO // FINANCIAL INFRASTRUCTURE

03 / THE CAPITAL RAIL

Connecting fiat capital to verified energy value: The capital rail operates as a bidirectional bridge between conventional monetary networks, productive energy assets, and verified Power Credits. It ensures compliant, contractually bound settlement between capital deployment and energy redemption.

FIAT CAPITAL	FINANCIAL RAILS	VERIFIED ENERGY	POWER CREDITS	FIAT SETTLEMENT
Investor or buyer deployment	Escrow & banking rails	Metered & audited kWh	Issued to wallet ledger	Contractual clearing & burn

FOUR COMMERCIAL SETTLEMENT USE CASES

USE CASE	MECHANISM & SETTLEMENT DESCRIPTION
PROJECT FINANCE	Fiat capital finances generation assets; metered energy triggers Power Credit issuance to funder wallets.
COMMERCIAL PURCHASE	Off-takers purchase Power Credits with fiat under contractual pricing, securing verified energy capacity.
PERFORMANCE-LINKED FINANCE	Verified energy evidence releases milestone fiat disbursements automatically via smart-contract rules.
REDEMPTION & SETTLEMENT	Redemption of Power Credits against commercial claims triggers final fiat clearing and burns the credit.

Bidirectional Settlement Loop (Fiat ↔ Power Credits)

Deployment (Fiat → Power Credit): Conventional capital enters via regulated rails to fund/procure physical energy, issuing audited Power Credits upon verified output.
Redemption (Power Credit → Fiat): Commercial off-takers redeem Power Credits to execute contractual settlement in fiat, permanently retiring the underlying credits upon execution.

Price Volatility & Risk Absorption Framework

Fixed Off-Take (Consumer Hedging): Buyer trades spot-market downside for multi-year budget stability and guaranteed 24/7 power matching.
Spot Indexing / Liquidity Buffers: Floating arrangements adjust redemption values dynamically, while capital rail liquidity pools absorb extreme market spreads.

04 / COMMERCIAL INTELLIGENCE

From energy evidence to productive capacity: Power Credits create a verified energy record. KYE and KYP extend that record into a clearer understanding of assets, operations, and commercial performance.

KYE — KNOW YOUR ENERGY

Identity, source, availability, and performance of energy assets. What is generated, stored, supplied, or consumed? How reliable and efficient is the asset?

KYP — KNOW YOUR PRODUCTIVITY

What energy enables in productive activity. What output and utilisation does it support? How efficiently is energy converted into economic output?

CONNECTED COMMERCIAL ASSESSMENT FLOW

KYC / KYB
Who participates?
→
KYE
What energy?
→
KYP
What productivity?
→
COMMERCIAL INTELLIGENCE
Full commercial picture

CAPITAL ALLOCATION & DECISION SUPPORT

Knowing who participates in the economy is important. Understanding what their assets and businesses can reliably produce makes investment decisions more informed. Commercial Intelligence supports benchmarking, underwriting, project evaluation, and performance monitoring.

05 / RESPONSIBLE INVESTMENT

Economic and social responsibility, measured: Responsibility is reflected in where capital is deployed, what it enables, and how outcomes are measured—not simply in stated intentions.

PILLAR	IMPACT OBJECTIVE & MEASUREMENT RULE
ECONOMIC	Direct capital towards productive assets, infrastructure, businesses, and measurable activity.
COMMERCIAL	Improve visibility of performance, utilisation, operating efficiency, and financial viability.
SOCIAL	Support reliable energy access, productive opportunities, skills, employment, and local enterprise.
ENVIRONMENTAL	Measure eligible efficiency and emissions-related outcomes where reliable data is available.
INSTITUTIONAL	Strengthen transparency, accountability, auditability, and confidence among investors and public institutions.

GOVERNANCE KEEPS THE SYSTEM SOUND

- **EVIDENCE:** Meter & asset data.
- **VERIFICATION:** Independent checks.
- **AUTHORITY:** Separated permissions.
- **LEDGER:** Traceable records.
- **SETTLEMENT:** Permitted payment rails.

MEASURABLE OUTCOMES

Each credit is traceable to its originating project, authorised issuance conditions, verified quantity, current owner, transfer history, and retirement status.